

Cybernetics Oriented Programming (CYBOP)

GNU GPLv3

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Application knowledge

CYBOL

Control software

CYBOI

Operating system

Hardware

Computer

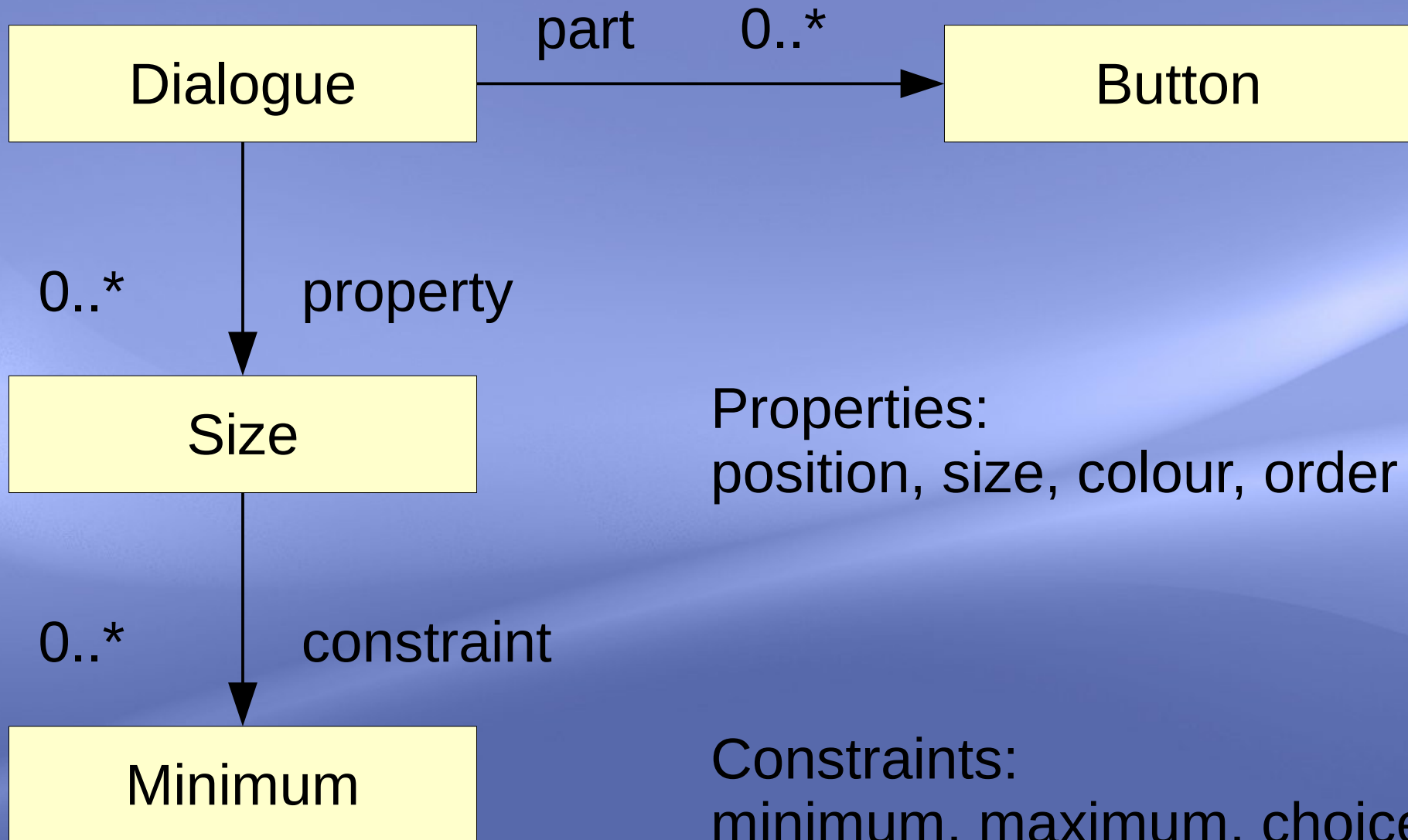
cyboi app/run.cybol

Criterion	Java world	CYBOP world
Language	Java	CYBOL (XML-based)
Interpreter	Java Virtual Machine (JVM)	CYBOI (written in C)
Theory	Object Oriented Programming (OOP)	CYBOP (knowledge schema)

Java Class

```
public class Dialogue {  
  
    // Structural attributes  
  
    private String title;  
    private Toolbar toolbar;  
    private Button button;  
  
    // Additional properties  
  
    private Point position;  
    private Dimension size;  
    private Color background;  
  
    // Methods  
  
    public void initialise(Object o);  
    public void process(Event e);  
}
```

Parts make up a whole



Properties:
position, size, colour, order

Constraints:
minimum, maximum, choice

State Template "dialogue.cybol"

```
<model>
  <part name="title" channel="inline" format="character" model="Test"/>
  <part name="toolbar" channel="file" format="element/part" model="path/file.cybol"/>
  <part name="button" channel="file" format="element/part" model="path/button.cybol">
    <property name="position" channel="inline" format="number/integer" model="5,5"/>
    <property name="size" channel="inline" format="number/integer" model="600,400"/>
    <property name="background" channel="inline" format="colour/terminal" model="red"/>
  </part>
</model>
```

Logic Template "initialise.cybol"

```
<model>
  <part name="build_menu" channel="inline" format="path/knowledge" model=".logic.menu"/>
  <part name="build_panel" channel="inline" format="path/knowledge" model=".logic.panel"/>
  <part name="display" channel="inline" format="communicate/send" model="">
    <property name="channel" channel="inline" format="meta/channel" model="terminal"/>
    <property name="encoding" channel="inline" format="meta/encoding" model="utf-8"/>
    <property name="language" channel="inline" format="meta/language" model="message/tui"/>
    <property name="format" channel="inline" format="meta/format" model="element/part"/>
    <property name="message" channel="inline" format="path/knowledge" model=".dialogue"/>
    <property name="clear" channel="inline" format="logicvalue/boolean" model="true"/>
  </part>
</model>
```

knowledge

view

partview

partview

knowledge.view.partview

logic

partlogic

knowledge.domain.partdo
main

partlogic

domain

partdomain

partdomain

runtime model hierarchy
with logic accessing states

is compound

name

abstraction

model

details

compound
knows how
its parts
discrete items
parts can be
compounds
= schema
or primitives

indicated by
abstraction

1..1

1..1

0..*

0..*

items may
belong to
a category

relations
unidirectional

traditional

cybop

**programme
structure**

**runtime
structure**



traditional



cybop

models suffer from
complexity
strong coupling /
dependencies
inflexible
difficult to maintain

one schema as memory
structure
directed acyclic graph (tree)
easily extensible
long-life software system

declarative language cybol
programming theory cybop
interpreter cyboi

flexible and adaptable at runtime
easy transparent communication

prototype-based programming similar to JavaScript

application creation by domain experts without programming knowledge

identical syntax for state (structure) and logic (function)
uniform usage of elements/commands

oriented on human thinking (which OOP only partly is)
double hierarchy separating micro-/macrocosm structure from meta properties
name „cybernetics“, because many of its original ideas were taken from nature

all knowledge hierarchical in one application tree
comparable to representation of website as DOM tree

cybol: just 4 tags and 4 attributes
unnecessary data are just ignored without error

bundling of attributes and methods – coupling, no flexibility
reflective meta architectures – bidirectional dependencies
bidirectional dependencies – complexity, circular references
global / static data access – untraceable data manipulation
container inheritance – falsified container content

problems

inter-disciplinary ideas

biology – configuration by dna

psychology – item + category + compound

physics – space & time

philosophy – body & mind

characteristics

long-life software system due to strict separation of application knowledge and system software

less dependencies due to representation of structural- and meta data in form of a double hierarchy

flexible architecture due to unified schema as the only remaining static memory structure

clean knowledge model code due to tree structure with unidirectional relations

universal communication due to star-like (not layer-like) translation architecture

easy handling due to one syntax describing states (space) and algorithms (time) in the same way

superfluous implementation phase due to direct interpretation of knowledge

solved problems

falsified container content due to container inheritance

fragile base class problem due to polymorphism

untraceable data manipulation due to global data access

circular references due to bidirectional dependencies

bidirectional dependencies due to reflective meta architectures

strong coupling due to bundling of attributes and methods

difficult maintenance due to high complexity



although cybop is free / open source software (GPL), it may well be used for commercial applications, since enterprise knowledge remains private

cybop is not in research for very much longer – discover its potential !

<http://www.cybop.org/>